

Ethan R Sampson

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EDUCATION

Master of Science in Bioinformatics

Northeastern University, Boston, MA.

Concentration in Omics and Data Analysis

Coursework: Bioinformatics Programming (Python), Bioinformatics Statistics (R programming), Computational Bioinformatics (Shell scripting and Perl), Genetics in Bioinformatics, Collecting and Storing Data (R programming), Bioethics, Transcriptomics, Proteomics, Machine Learning

Grad. May 2025

GPA: 3.4/4.0

Bachelor of Science in Biology

Loyola University Maryland, Baltimore, MD.

Athletics: Division 1 Varsity Swimming for The Greyhounds.

Coursework: Cancer Biology, Genetics, Pharmacology, Organic Chemistry, Biochemistry, Statistics, Calculus, General Biology

Grad. May 2023

GPA: 3.0/4.0

TECHNICAL SKILLS

Soft Skills: Communication, Teamwork, Leadership, Problem-Solving, Positive, Adaptable, Creative, Time-management, organization

Programming Languages: Python, R, Shell, Perl, SQL

Bioinformatic Tools: BLAST, Integrated Genome Viewer (IGV), Bioconductor, NCBI, FASTA, BioPerl, QUAST, MOE

Laboratory Techniques: Western Blots, Cell Viability Assays, PCR, Gel electrophoresis, Buffer preparation, Lab ELN

BIOINFORMATIC / DATA SCIENCE SKILLS

R: Used for statistical analysis of genomic data, data wrangling, and building machine learning models with model evaluation.

SQL: Used SQLite to organize and query biological datasets; applied joins, aggregations, and filtering for efficient data manipulation.

Python: Creating programs that work with genetic and other biological data. This includes handling, organizing, and creating bioinformatics files, sequencing genetic data, creating contigs, and creating coverage analysis.

WORK EXPERIENCE

Downstream Process Bioinformatics Co-op

January 2025 – June 2025

Pfizer BioProcess Research and Development, Andover MA

- Create a biomolecular descriptor database from amino acid sequences (proteomic data) using MOE, Python, and Excel
- Utilized data mining and analysis skills to build the database
- Documented workflows for reproducibility, collaborated cross-functionally, and met deadlines independently

ACADEMIC PROJECTS

Northeastern University

September 2023 – May 2025

- Led collaborative bioinformatics projects, applying data mining, machine learning, and NGS data analysis techniques
- Built predictive models including logistic regression, random forests, and k-nearest neighbors; used cross-validation and performance metrics to evaluate models.
- Performed quality control (FastQC, MultiQC), read alignment, and quantification of transcriptomic data.
- Conducted differential expression analysis using DESeq2 and visualized results with ggplot2 and heatmaps.
- Applied functional annotation and gene ontology (GO) analysis to interpret biological significance.
- Applied supervised machine learning models (logistic regression, random forest) on biological datasets to classify phenotypes and assess model accuracy using confusion matrices and ROC curves.

RESEARCH EXPERIENCE

Examining Protein Expression in Cancer Cells

January 2023 – May 2023

Undergraduate Student Researcher, Loyola University

- Performed independent Western blot experiments
- Quantified protein bands and visualized results using Excel and R
- Created and performed a presentation on results. As well as creating a research paper on the results.